

Novelty Summary

The proposed Digital Twin Cloud Platform for Smart Campus Sustainability Analytics offers a unified method for monitoring, analyzing, and improving campus sustainability. Unlike systems that focus on specific areas like energy monitoring, indoor comfort, BIM visualization, or facility management, this platform merges various sustainability factors with cloud-based Digital Twin technology, predictive analytics, and decision-support tools.

New Feature: A dynamic Campus Sustainability Score will assess individual campus buildings based on energy consumption, water usage, occupancy, environmental conditions, and estimated carbon emissions. A what-if simulation feature will enable administrators to explore the effects of operational changes before putting them into practice on the campus.

Better Algorithm: Predictive analytics will forecast resource consumption and identify unusual usage patterns. The sustainability scoring method will include several normalized indicators rather than just assessing one resource like electricity.

Better Architecture: A modular Digital Twin design will connect data collection, cloud processing, analytics, prediction, simulation, and visualization. This structure will let multiple campus buildings and sustainability parameters be managed through a single platform.

Better AWS Integration: AWS services can be integrated for IoT data ingestion, cloud storage, serverless processing, machine-learning analytics, monitoring, authentication, and visualization, allowing different components of the Digital Twin to operate within a common cloud environment.

Better Security: Role-based access and secure cloud authentication will limit access to sustainability data and administrative tasks. Cloud security measures will also safeguard communication among data sources, processing elements, and the dashboard.

Better Automation: The platform will automatically handle incoming data, update Digital Twin states, calculate sustainability scores, detect abnormal consumption, generate alerts, and refresh analytics without needing constant manual oversight.

Better Accuracy: Instead of depending on a single sustainability parameter, the system will merge various operational and environmental indicators. Predictive models will also be assessed using suitable performance metrics before deployment to enhance the reliability of resource forecasts.

Better Scalability: The cloud-based framework allows for the addition of more buildings, sensors, sustainability parameters, and data sources without needing to redesign the entire system.